

Sustainability Org: A GenAI-Powered Signal Detection and Recommendation Engine

Project Work Final Presentation

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July 2026



Presentation Outline

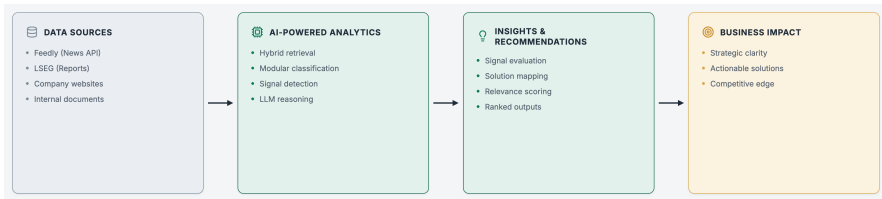
- ➊ Introduction
- ➋ Literature Survey / Background Theory
- ➌ Problem Definition
- ➍ Objectives
- ➎ Methodology
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- ➐ Conclusion & Future Scope
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- GenAI and Large Language Models have transformed how organizations extract meaning from unstructured text.
- **Sustainability Org** is a GenAI-powered decision-intelligence engine built during an internship at **Bain Capability Network**.
- It treats the problem as **signal detection**: finding latent business signals in fragmented evidence.
- Signals are **validated by retrieval-grounded LLM reasoning** and mapped to ranked, explainable recommendations.
- Designed to be **modular, explainable, and human-supervised**.



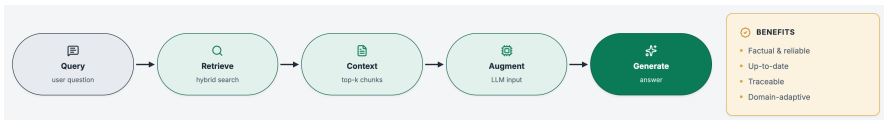
Introduction - System at a Glance



From trusted data sources, through AI-powered analytics and signal evaluation, to ranked recommendations that drive better-grounded decisions.



- **In-context learning:** LLMs can be steered for classification via prompts alone - enabling modular, YAML-configurable classification.
- **Retrieval-Augmented Generation (RAG):** grounds LLM reasoning in retrieved, citable evidence instead of parametric memory.
- **Hybrid retrieval:** fuses lexical (BM25) + dense (vector) search, then re-ranks with a cross-encoder for precision and semantic recall.



Problem Definition

- Organizations must interpret vast, fragmented, heterogeneous evidence to make decisions.
- Keyword search and rigid pipelines miss context and are hard to audit.
- Off-the-shelf LLM tools hallucinate and provide no provenance or oversight.

Goal

Transform heterogeneous, unstructured text into structured, actionable, and **explainable** intelligence - at scale, while keeping a human analyst in the loop.

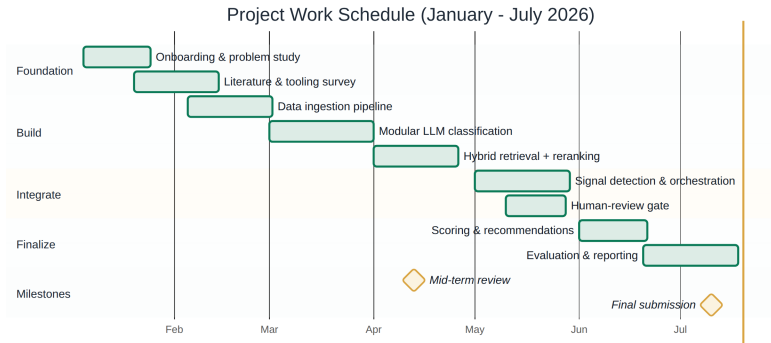


Objectives

- Build a scalable ingestion & normalization pipeline for heterogeneous sources.
- Design a modular, YAML-configurable LLM classification framework.
- Implement hybrid retrieval (BM25 + vector + cross-encoder reranking).
- Build a reasoning-driven signal-detection framework with agentic web search.
- Enforce a human-in-the-loop review gate before expensive processing.
- Map validated signals to solutions and produce weighted, ranked recommendations.



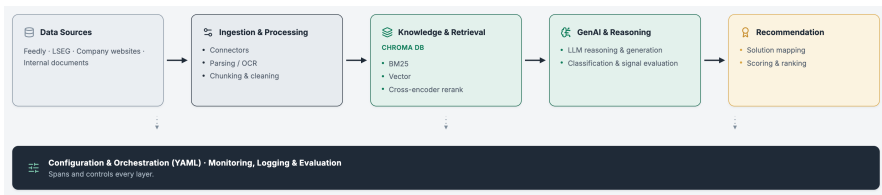
Project Timeline



Six-month internship (Jan–Jul 2026): overlapping phases from onboarding and ingestion through classification, retrieval, orchestration, and evaluation - with the mid-term review and final submission milestones.



Methodology - System Architecture

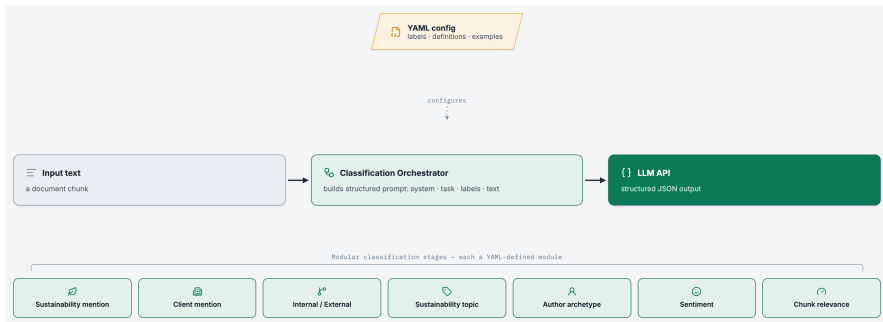


Five layers - ingestion, classification, hybrid retrieval, reasoning-based signal detection, and recommendation - all controlled end-to-end by YAML configuration.



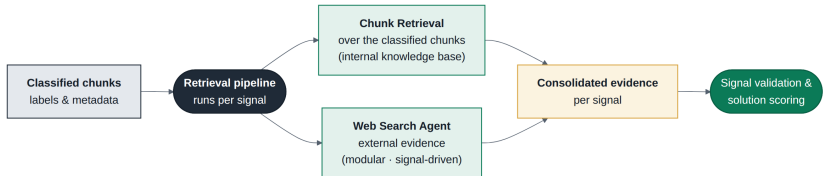
Methodology - Modular LLM Classification

- Each dimension is a configurable, prompt-driven module; executed in parallel over **Dask** partitions, each label carrying a confidence score.



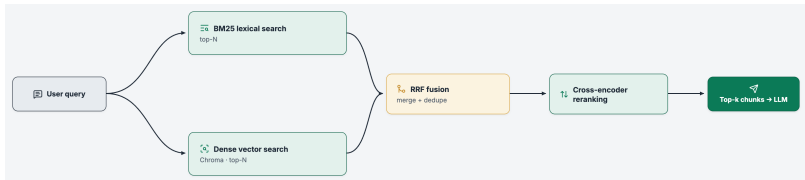
Methodology - Retrieval Pipeline (Gathering Evidence)

- Once classification is complete, the **retrieval pipeline** runs **per signal** to gather the evidence needed to validate it.
- Two complementary engines run for each signal: **chunk retrieval** over the internal knowledge base, and a **web search agent** for external evidence.
- Their outputs are consolidated per signal before validation and scoring.



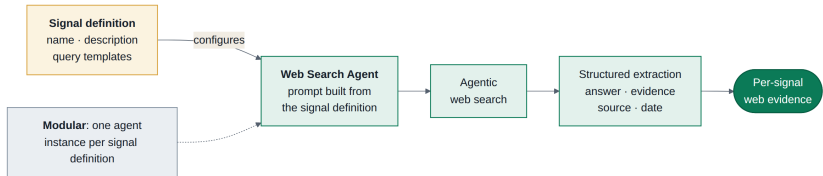
Methodology - Chunk Retrieval (over Classified Chunks)

- The **classified chunks** become the searchable knowledge base - chunk retrieval runs directly over the classification output.
- Retrieval is **scoped per signal** using metadata filters derived from the classification labels.
- Each query uses **hybrid search** - BM25 (lexical) + dense vector, fused via RRF - then **cross-encoder reranking** to return the top-*k* evidence chunks.

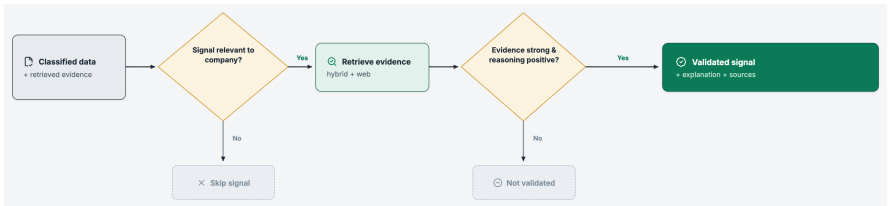


Methodology - Web Search Agent (Modular, Signal-Driven)

- Complements the internal knowledge base with **external web evidence** for each signal.
- **Modular by design**: one agent instance per signal, **configured from the signal's definition** - its name, description, and query templates.
- Runs an **agentic web search** and returns **structured evidence** - answer, supporting quote, source, and date - for each signal.



Methodology - Signal Detection & Human Review

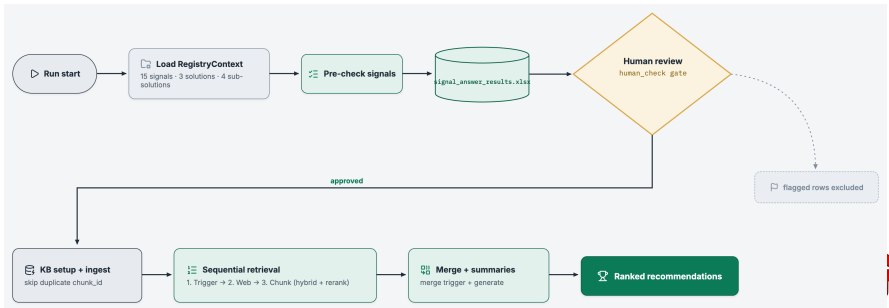


Pre-check → evidence retrieval → LLM evaluation → validation → output. A mandatory human_check review gate excludes weak candidates before costly retrieval.



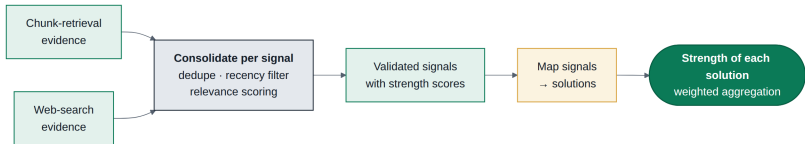
Methodology - Orchestration & Human-Review Gate

- **Registry-scoped:** a signal registry (catalogue) and a solution registry (active scope) drive the run.
- **Strictly sequential:** pre-check → review → KB ingest → trigger → web → chunk - no blind parallelization.
- **Human-in-the-loop:** the run **pauses** at the `human_check` gate before any expensive retrieval.



Methodology - Consolidation & Solution Strength

- Evidence from **chunk retrieval** and the **web search agent** is **consolidated per signal** - deduplicated, recency-filtered, and relevance-scored.
- Each signal is validated and assigned a **strength score** from its consolidated evidence.
- Signals are mapped to solutions, and a **weighted aggregation** computes the **strength (fit)** of each solution.



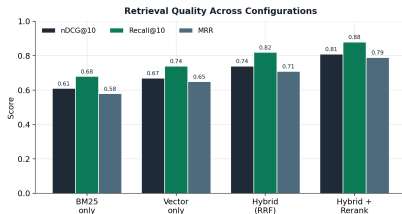
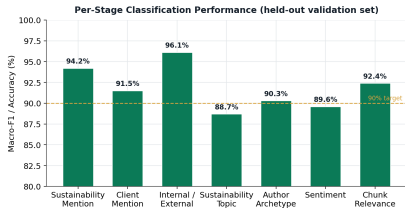
Methodology - Recommendation Engine



Validated signals are mapped to solutions and scored by a static-weightage engine, producing ranked, source-traceable recommendations.



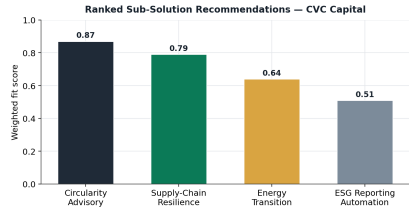
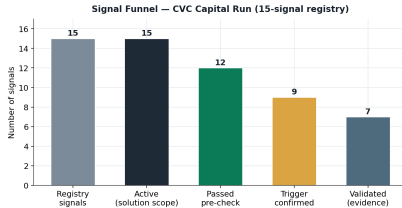
Result Analysis - Classification & Retrieval Quality



Classification mean macro-F1 $\approx$ **91%** (left). Hybrid + reranking lifts nDCG@10 from **0.61** $\rightarrow$ **0.81**, a $\approx$ 33% gain (right).



Result Analysis - Signal Funnel & Recommendations



CVC Capital run: 15 registry signals narrow to **7 validated** (left). “Circularity Advisory” is the best-fitting recommendation at 0.87 (right) - every score traces to validated signals and cited evidence.



Screenshots - Classification Pipeline (Notebook)

classification_flow_v1.ipynb

Python 3 — idle

+ Code ▷ Run ■ Stop ↺ Restart

```
[14]: # Stage 3-8: run modular LLM classification across Dask partitions
from src.helper_functions import classification_helper_functions as clf
ctx = clf.load_user_config("config/user_instructions.yaml")
df = clf.final_pipeline_to_run_along_dask(raw_df, ctx, npartitions=8)
```

```
[14]: [config] company_in_focus = 'CVC Capital'
[dask] LocalCluster(8 workers) ... ready
[stage] sustainability_mention ██████████ 4120/4120 38.2s
[stage] client_mention ██████████ 4120/4120 41.0s
[stage] sustainability_topic ██████████ 4120/4120 46.9s
[stage] sentiment ██████████ 4120/4120 27.4s
✓ classification complete - 4120 rows · mean confidence 0.883
```

```
[15]:
```

	doc_id	sust_mention	topic	sentiment	confidence
0	FEEDLY_5521	Yes	Energy Transition	Positive	0.94
1	LSEG_0187	Yes	Circularity	Neutral	0.87
2	WEB_2231	Yes	Emissions	Negative	0.79
3	LSEG_0188	Yes	Supply Chain	Positive	0.96
4	FEEDLY_5525	Yes	Governance	Neutral	0.85

Six modular LLM stages run across 8 Dask partitions over 4,120 rows (mean confidence 0.883), producing the labelled dataframe.



Screenshots - Web-Retrieval Results (Excel)

Excel web_retrieval_results_cvc.xlsx						
D2 fx Yes – CVC Capital committed to a 2040 net-zero portfolio target						
	A	B	C	D	E	F
1	signal_id	signal_name	answer	evidence_summary	confidence	human_check
2	SIG_01	Net-Zero Commitment	Yes	2040 net-zero portfolio target across EU buyout funds.	0.94	1
3	SIG_03	Circular Economy Focus	Yes	Two holdings scaling closed-loop packaging.	0.88	1
4	SIG_04	Supply-Chain Exposure	Yes	Elevated Scope-3 exposure in two industrials.	0.81	1
5	SIG_07	Regulatory Pressure (CSRD)	Yes	CSRD reporting applies from FY2025 for EU holdings.	0.86	1
6	SIG_09	Green Financing Activity	No	No sustainability-linked loan issuance found.	0.72	0
7	SIG_12	Emissions Disclosure Gap	Yes	Scope-3 methodology absent for 4 of 11 holdings.	0.78	1
8	SIG_14	Renewable Investment	Yes	Fresh capital into grid-scale storage platform.	0.85	1
signal_answers trigger_answers results registry_scope						

Each signal answer is grounded in a cited source with a confidence score and the analyst's `human_check` decision.



Screenshots - Orchestration Run (Terminal)

```
analyst@sustainability-org — orchestration run

$ python -m src.run_orchestration --stage all --company "CVC Capital"

[12:41:02] registry RegistryContext — signals=15 · solutions=3 · sub-solutions=4
[12:41:02] registry stage split — pre-check=3 · trigger=2 · scalar=10
[12:41:03] precheck wrote signal_answer_results.xlsx — 12 rows
[12:41:03] review ■ human_check gate — awaiting reviewed workbook
[12:48:20] review ► resumed — 2 rows excluded (SIG_09, SIG_15)
[12:48:24] kb_ingest ingested 1,846 chunks (skipped 213 dup) ✓
[12:48:24] trigger 2 signals — SIG_07 ✓ SIG_10 ✓ 18.9s
[12:48:43] web agentic retrieval — 10 scalar × 4 entities
[12:51:59] web signal_answers=41 · results=63 · failures=0 196.4s
[12:52:00] chunk hybrid (BM25+dense) + rerank — top_k=8
[12:52:32] chunk evidence rows=214 32.1s
[12:52:47] summary signal-level=7 · sub-solution=4 · ranked recs written
[12:52:47] ✓ run complete — validated signals=7/15 · wall-clock=4m 21s

$
```

A complete strictly-sequential run with the human_check gate: 15 signals → 7 validated in $\approx 4\text{m } 21\text{s}$.



Conclusion

- A modular, retrieval-grounded, human-supervised GenAI engine reliably turns unstructured evidence into ranked, explainable recommendations.
- Met targets: $\approx 91\%$ classification F1, nDCG@10 of 0.81, 7 validated signals, fully source-traceable outputs.

Future Scope

- Real-time, incremental (upsert-based) ingestion and continuous monitoring.
- Learned / feedback-calibrated scoring and larger gold-standard evaluation.
- Interactive dashboards and generalization beyond sustainability.



- ❶ Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *NeurIPS*, 2020.
- ❷ Brown et al., “Language Models are Few-Shot Learners,” *NeurIPS*, 2020.
- ❸ Robertson & Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *FnTIR*, 2009.
- ❹ Karpukhin et al., “Dense Passage Retrieval for Open-Domain QA,” *EMNLP*, 2020.
- ❺ Nogueira & Cho, “Passage Re-ranking with BERT,” *arXiv:1901.04085*, 2019.
- ❻ Cormack et al., “Reciprocal Rank Fusion,” *ACM SIGIR*, 2009.
- ❼ Manning et al., *Introduction to Information Retrieval*, Cambridge Univ. Press, 2008.



Thank You

